

Maximum Likelihood Estimation

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Maximum Likelihood Estimation

- Problem Definition

- Given a dataset $\mathcal{D} = \{X_1, X_2, \dots, X_n\}$ drawn from a distribution $P(X; \theta)$, i.e., $X_i \sim P, \forall i$
- Assumptions:
 - The samples are independent and identically distributed (i.i.d.)
 - P is known but not the parameters θ
- Aim: Estimate θ

- Maximum Likelihood Estimation

- The likelihood function:

$$l(\theta|\mathcal{D}) = P(\mathcal{D}; \theta) = \prod_{i=1}^n P(X_i; \theta)$$

- For computational purpose, we consider log-likelihood:

$$ll(\theta|\mathcal{D}) = \sum_{i=1}^n \log P(X_i; \theta)$$

Maximum Likelihood Estimation

- The maximum likelihood estimate of θ :

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} \quad ll(\theta|\mathcal{D})$$

- We can take partial derivatives (of $ll(\theta|\mathcal{D})$) and equate to zero and find the closed form solution if possible
- If no closed form solution, iterative methods can be used

Problems

Estimate the parameters. Find the bias and variance of the estimates.

Let $X_1, X_2, \dots, X_n$ be i.i.d. samples drawn from

- ① $\mathcal{N}(\mu, \sigma^2)$ where μ is known and σ^2 is unknown. Derive the ML estimate of σ^2 .
- ② $\mathcal{N}(\mu, \sigma^2)$ where both μ and σ^2 are unknown. Derive the ML estimate of μ, σ^2 .
- ③ $\text{Bernoulli}(\theta)$. Derive the ML estimate of θ .
- ④ $\text{Uniform}([a, b])$ for $a, b \in \mathbb{R}$. Derive the ML estimate of a and b .
- ⑤ $\text{Exp}(\lambda)$. Derive the ML estimate of λ .
- ⑥ from $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$, for some $\boldsymbol{\mu} \in \mathbb{R}^d$ and some $\Sigma \in \mathbb{R}^{d \times d}$. Derive the ML estimate of $\boldsymbol{\mu}, \Sigma$.

Maximum Likelihood Estimation (Independent but not Identical)

- Suppose the samples $\mathcal{D} = \{X_1, X_2, \dots, X_n\}$ are independent but not identical
- $P_1, P_2, \dots, P_n$ are known but not the parameters
- **Problem:** Let $X_1, X_2, \dots, X_n$ be independent real valued random variables, where each X_i is distributed as $\mathcal{N}(\mu_i, \sigma^2)$. Here $\mu_1, \mu_2, \mu_3, \dots, \mu_n$ are known constants and σ^2 is the unknown parameter.
 - Write the loglikelihood function, $l(\theta|\mathcal{D})$.
 - Derive the ML estimate of the parameter σ^2 .
 - Find the bias and variance of the estimate.